

Sea Level Rise Global Analysis

Big Data & Cloud Computing for Climate

Context

Why should we be concerned about sea level rising?

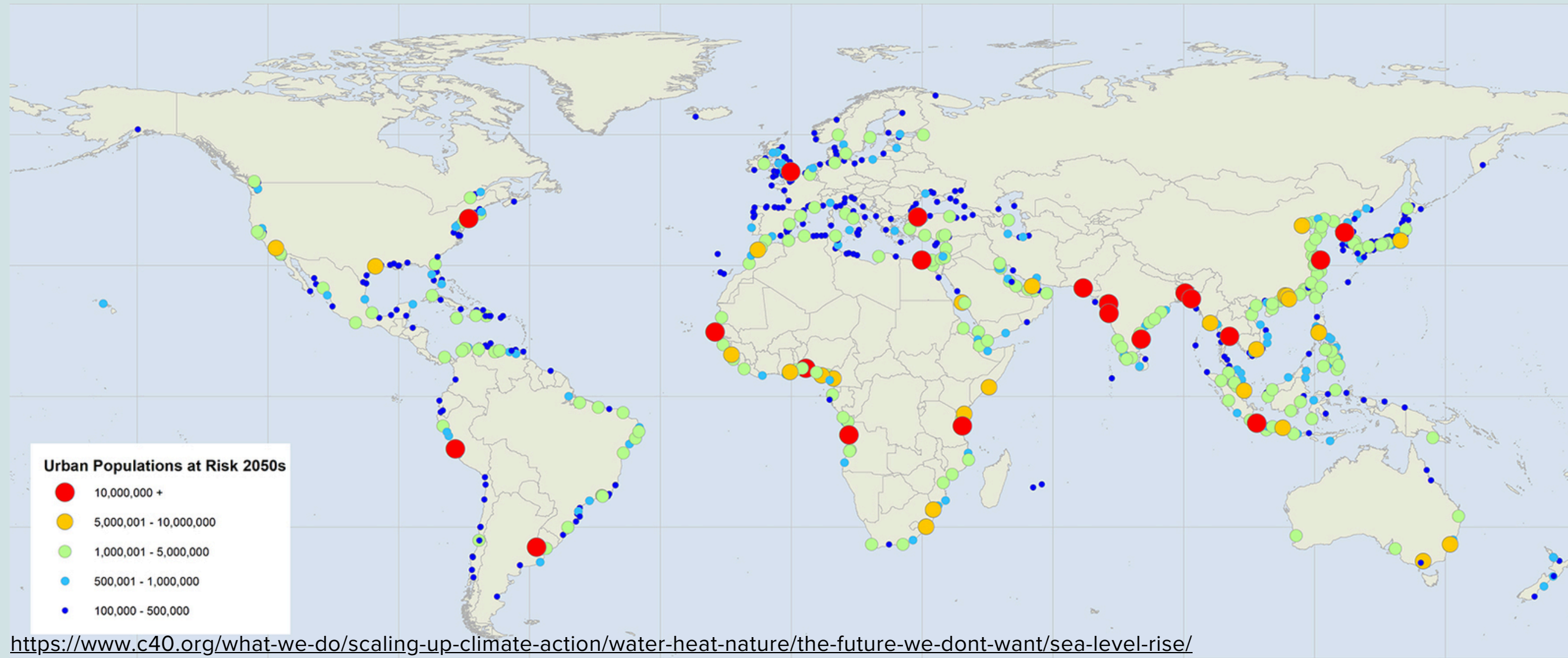
It threatens coastal communities, ecosystems, and global economies, putting millions at risk. Acting now can help mitigate these impacts and protect future generations.

Level rising drivers

- Sea Ice melting
- Thermal expansion of sea water

Our dataset

18 years of altimetry data between 01/1998 - 01/2016 (space-time dimensions, ssh, sst)



Context

How is SSH rising measured?

Sea Surface Height (SSH) rise is measured by observing changes in sea level over time using satellite altimetry and tide gauges.

With which instruments?

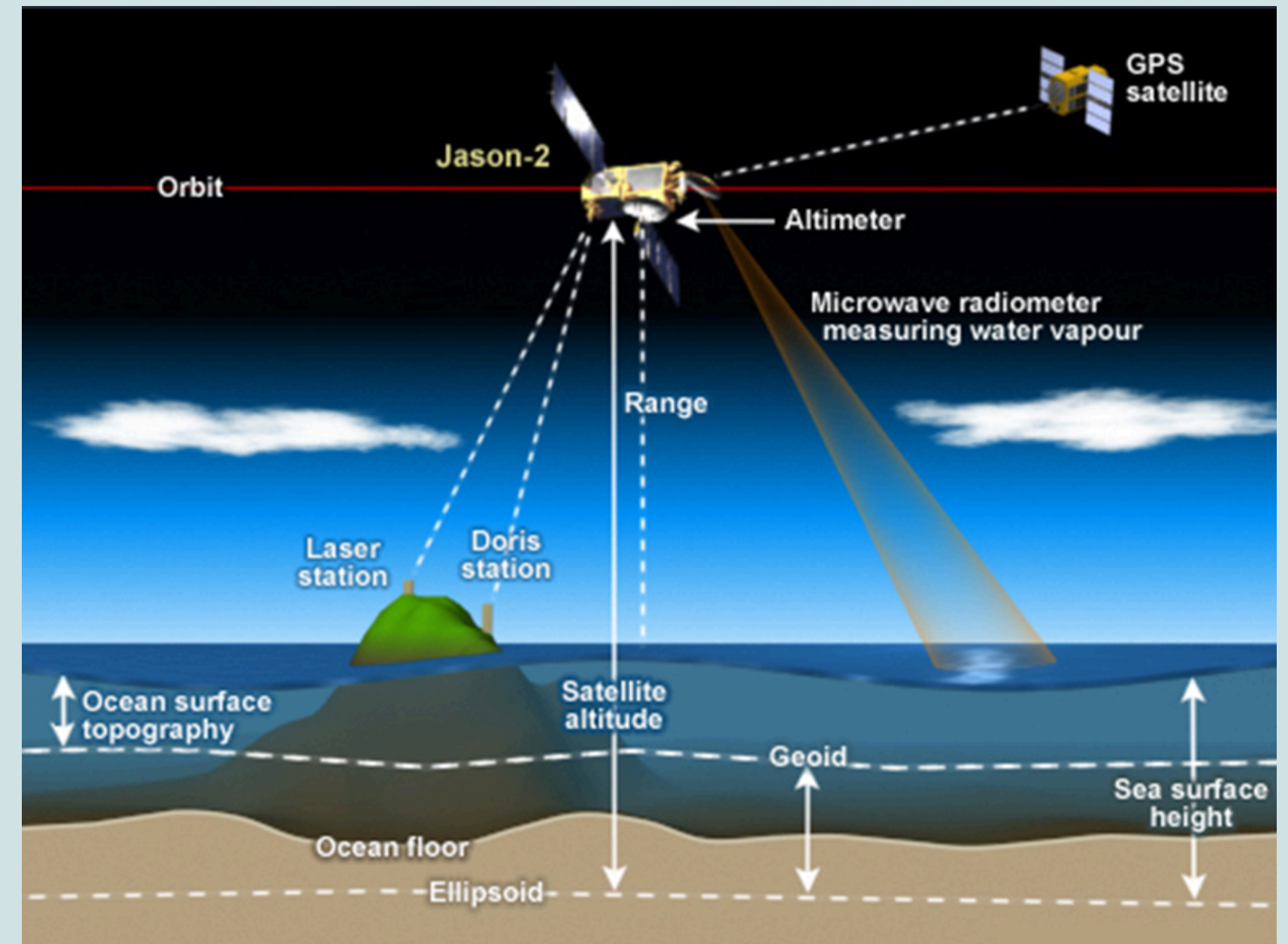
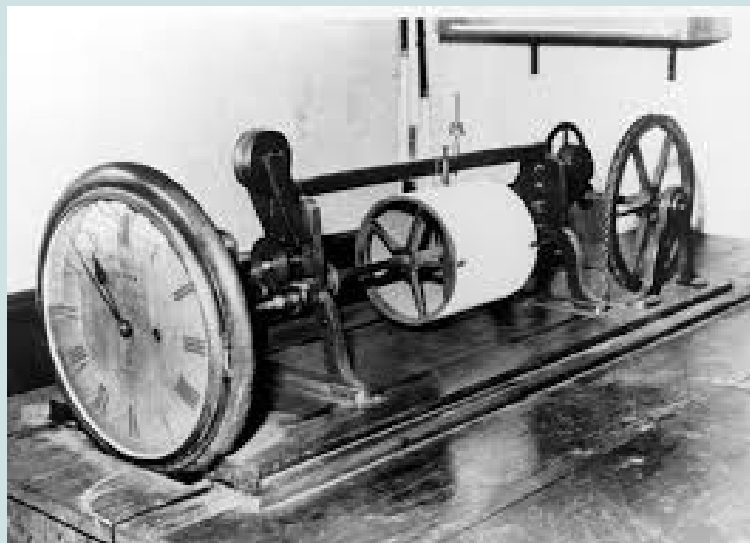
Satellite altimeters (e.g., TOPEX/Poseidon, Jason series, Sentinel-6)

Tide gauges (coastal stations)

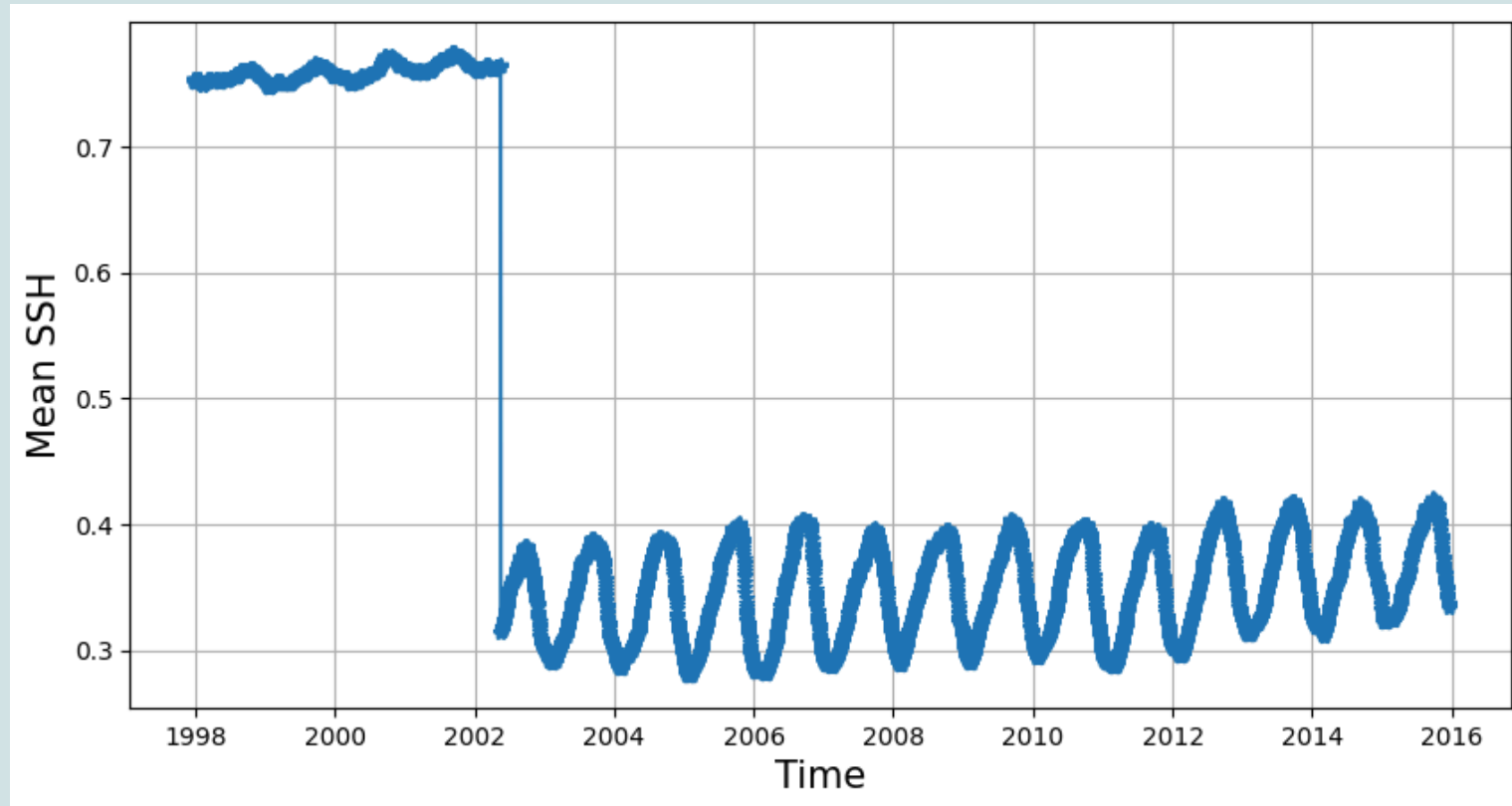
Since when?

Tide gauges: Early 19th century (~1807 onwards)

Satellite altimetry: Since 1992 (TOPEX/Poseidon mission)



The dataset



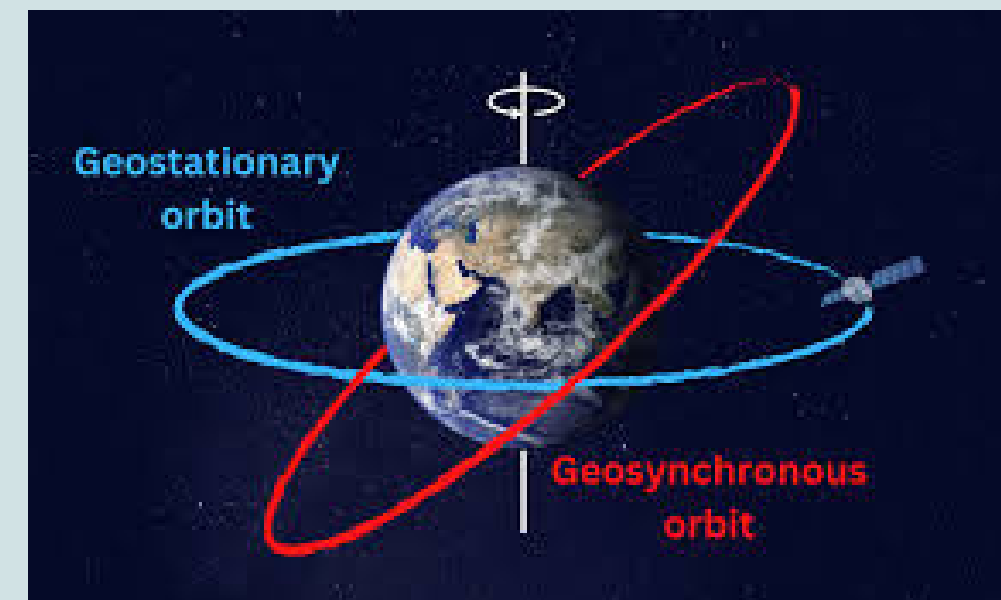
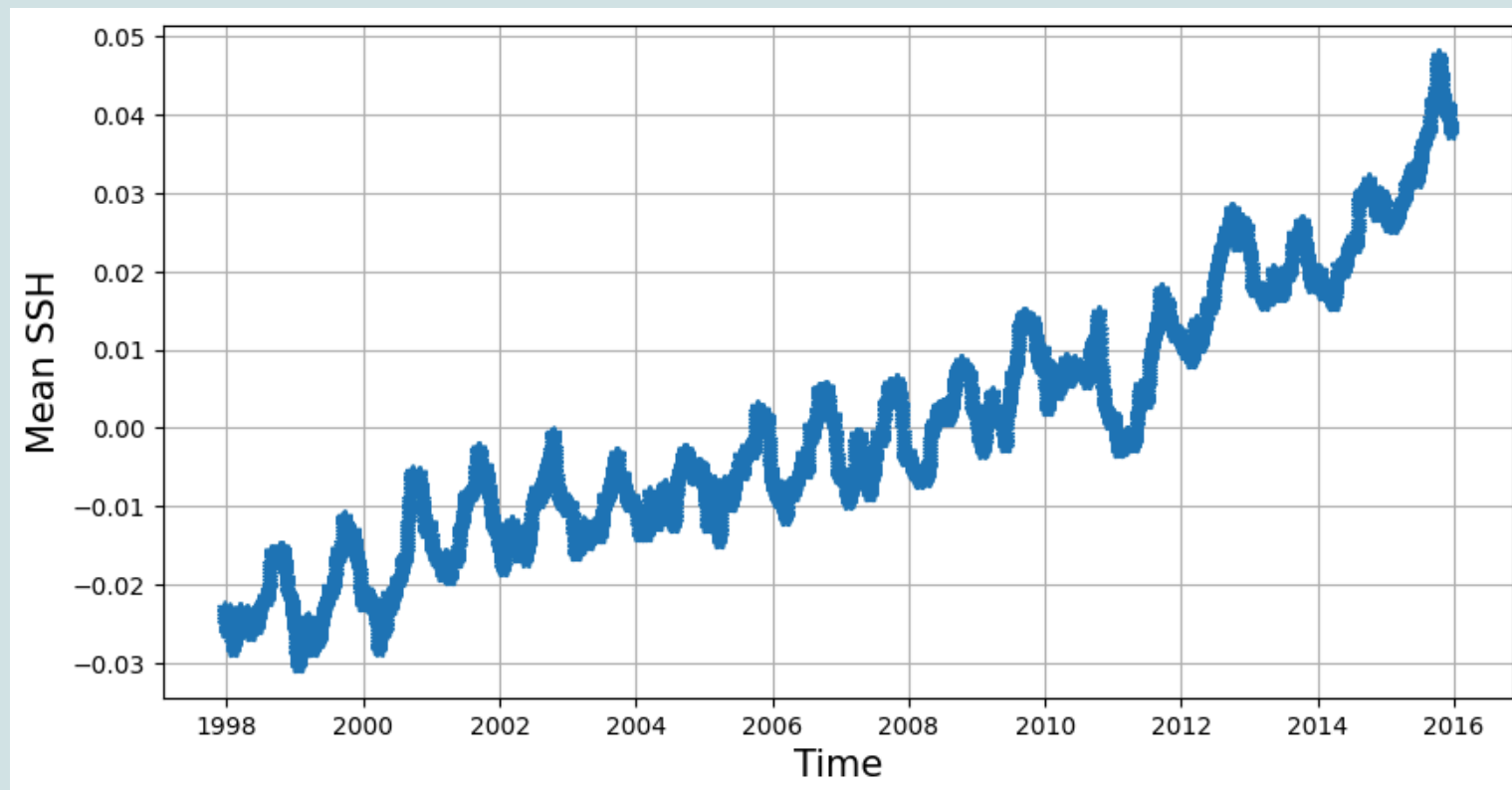
Our dataset: gap in 15th May 2002

==> Launch of Haiyang-1A

==> Not the same inclination

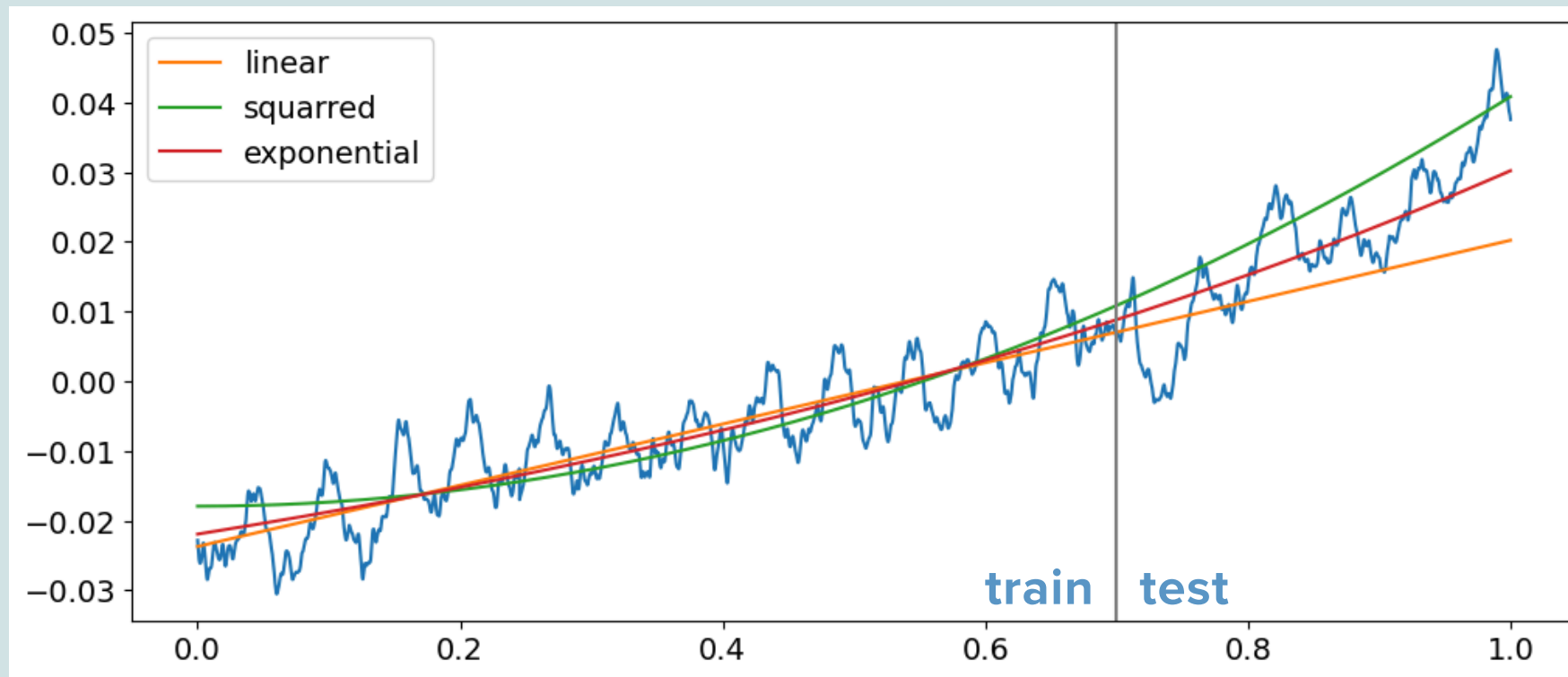
==> Not the same global coverage

==> Not the same mean SSH

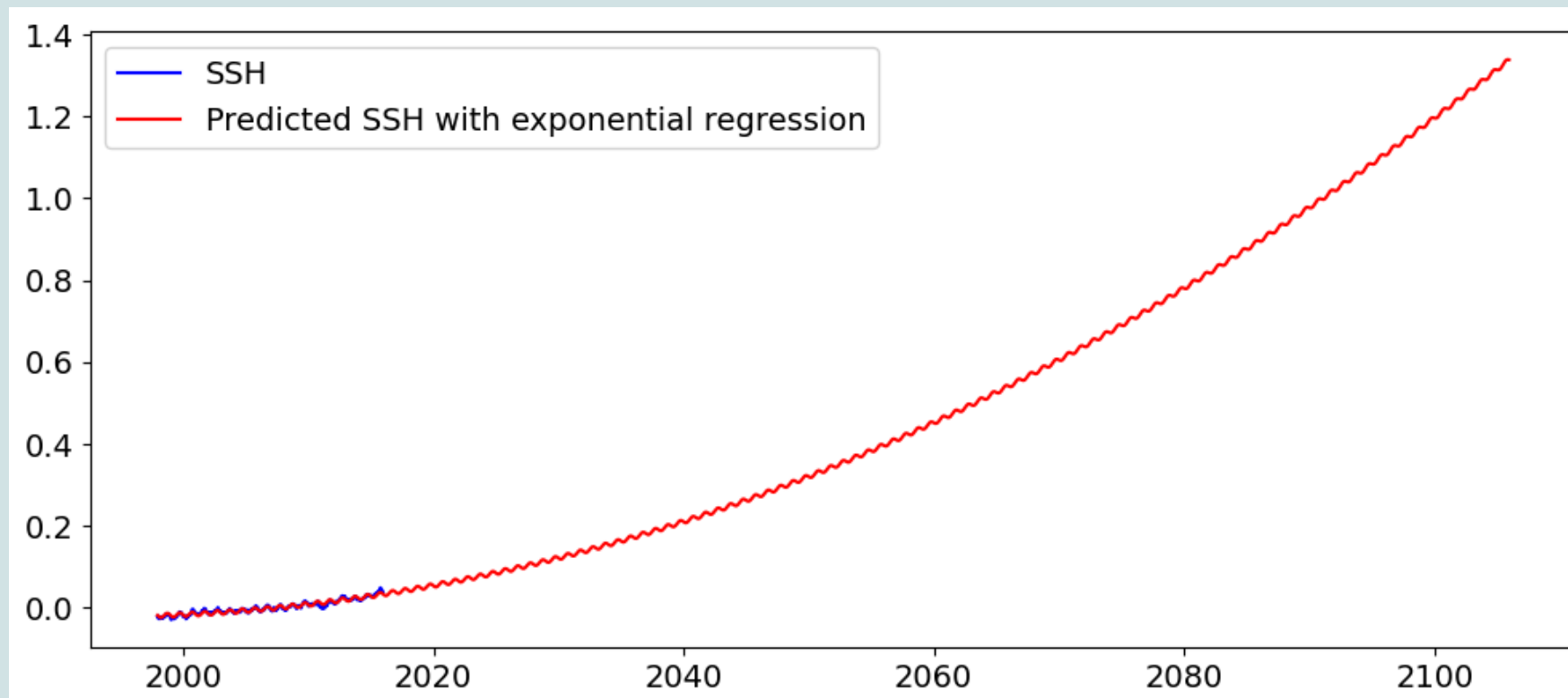


Focus between -40 and +40 latitudes

Our Model



The global trend: the exponential trend (combined with a linear coefficient) seems to be the best, with the highest R^2 .

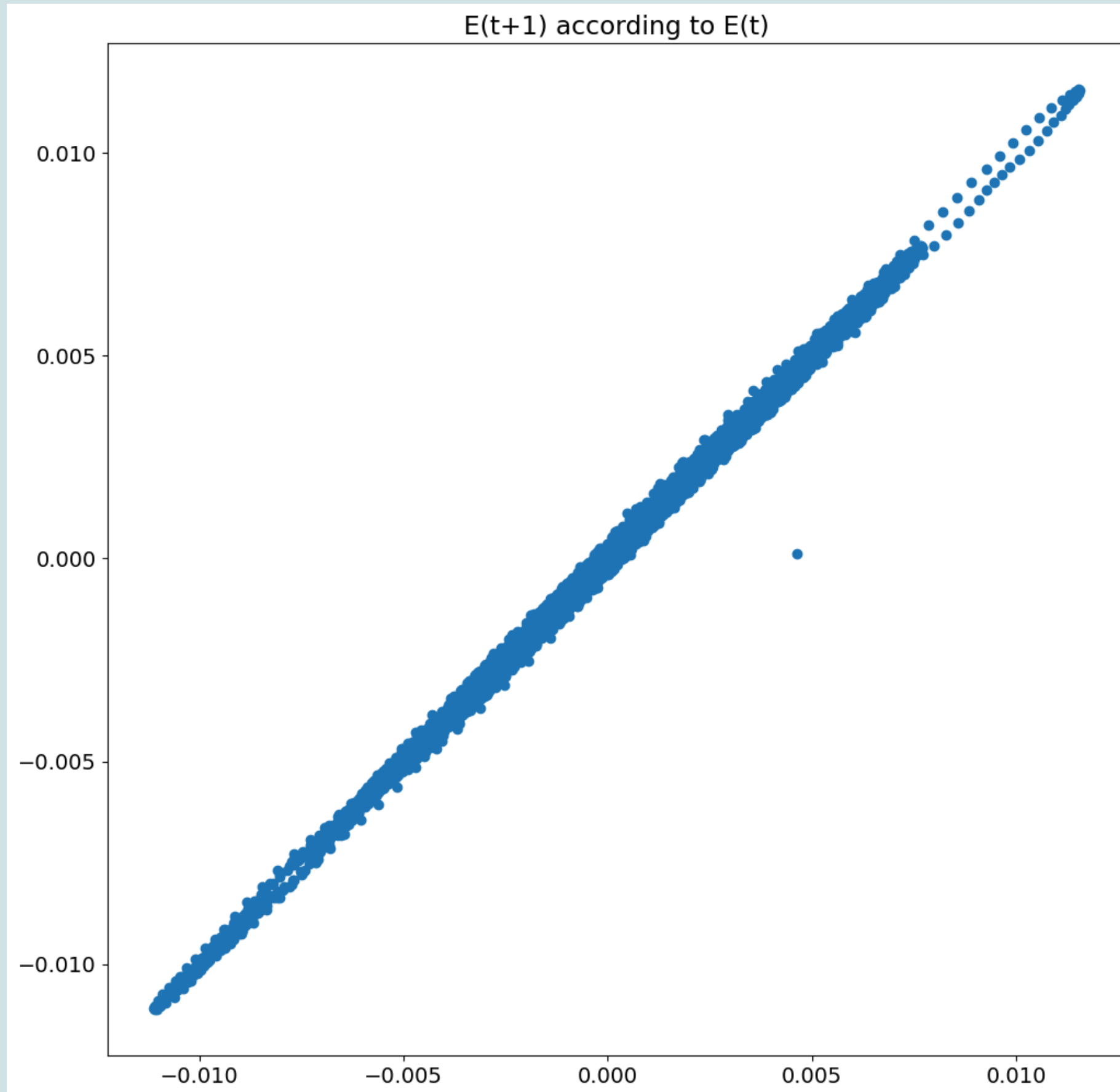


Linear regression with the following features:

- t
- $e^{\frac{t}{t_0}}$
- $\cos(\omega t)$ with $\omega = \frac{2\pi}{365.25}$
- $\sin(\omega t)$

Target: mean_ssh in the area $-40 < \text{lat} < 40$

Autoregression



residuals = predicted values - observed values

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (residuals)^2$$

with N the number of available times in the data

$Slope = 0.9989 \simeq 1 \rightarrow$ **Linear** model for the prediction using the variance

So the prediction of uncertainty at time t is

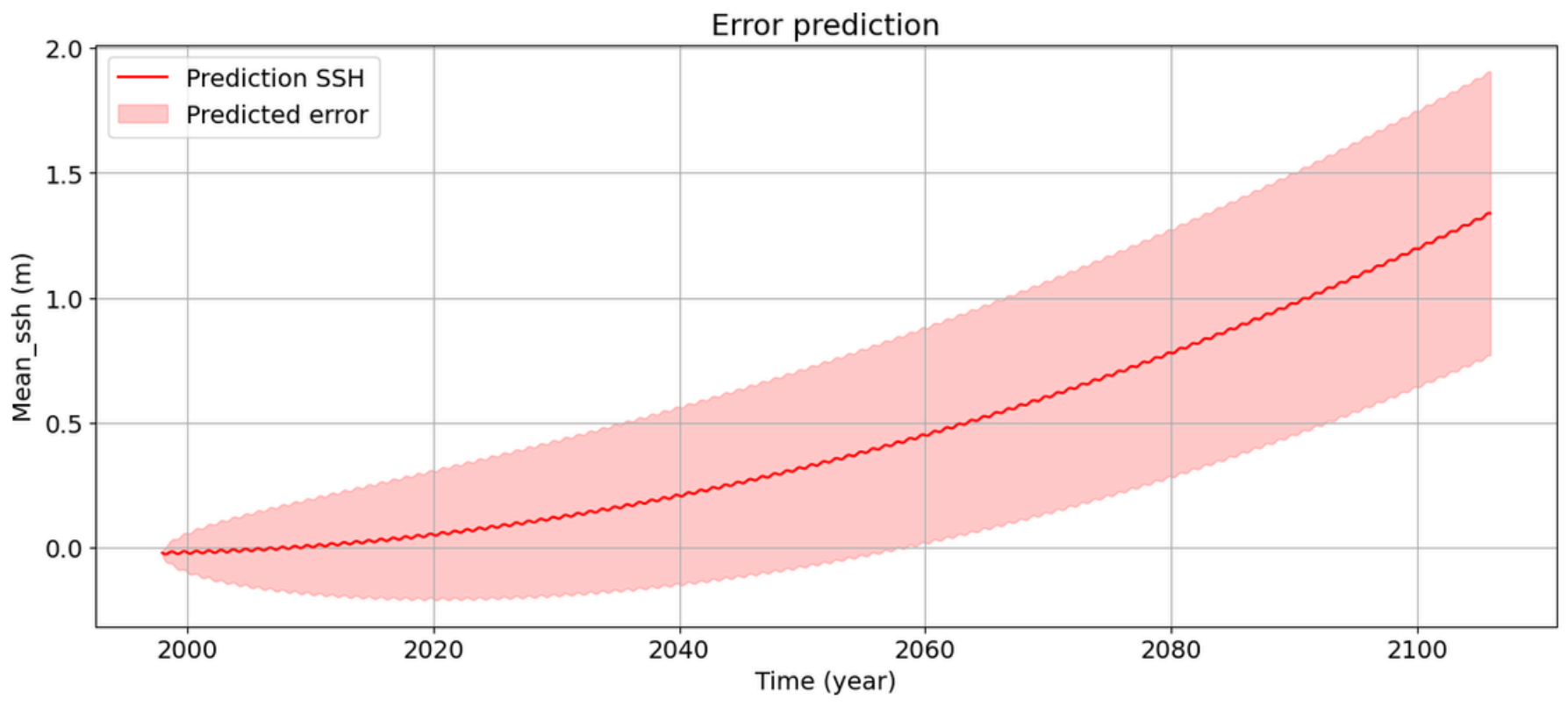
$$Var(\varepsilon(t)) = t * \sigma^2$$

So the function for the error will be :

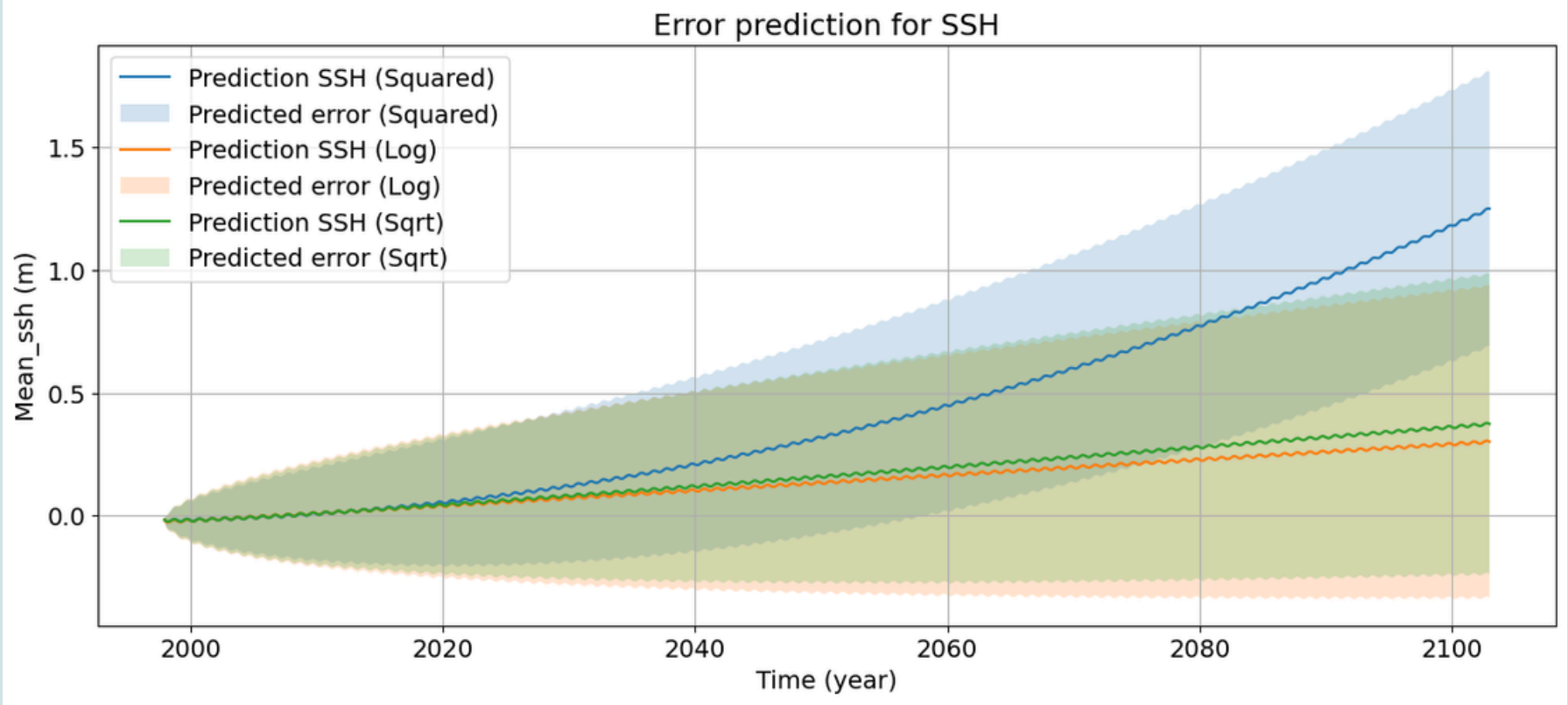
$$std(\varepsilon(t)) = \sqrt{t} * \sigma$$

The error in time “t+1” according to the error in time “t”

Prediction with 68.2 % confidence interval

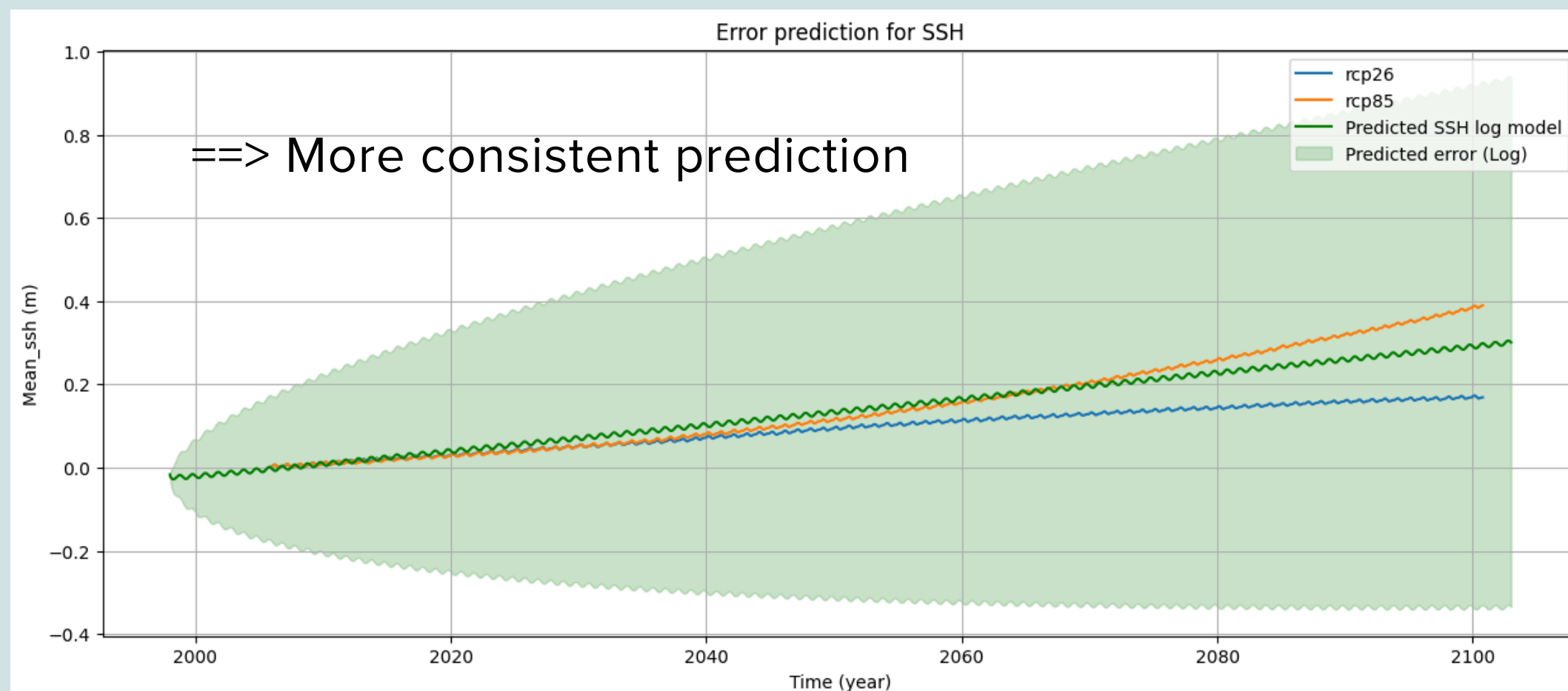
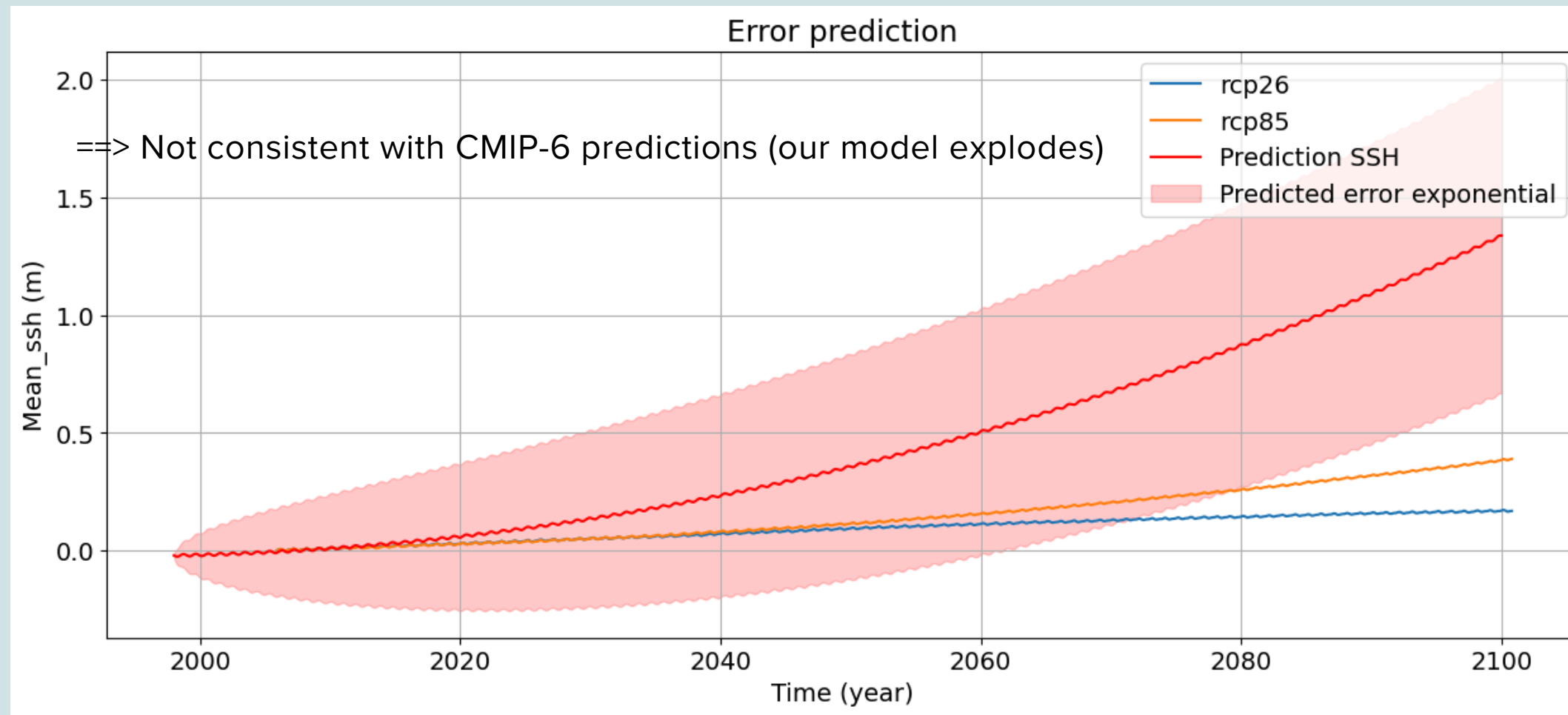


Exponential



Squared
Logarithmic
Square root

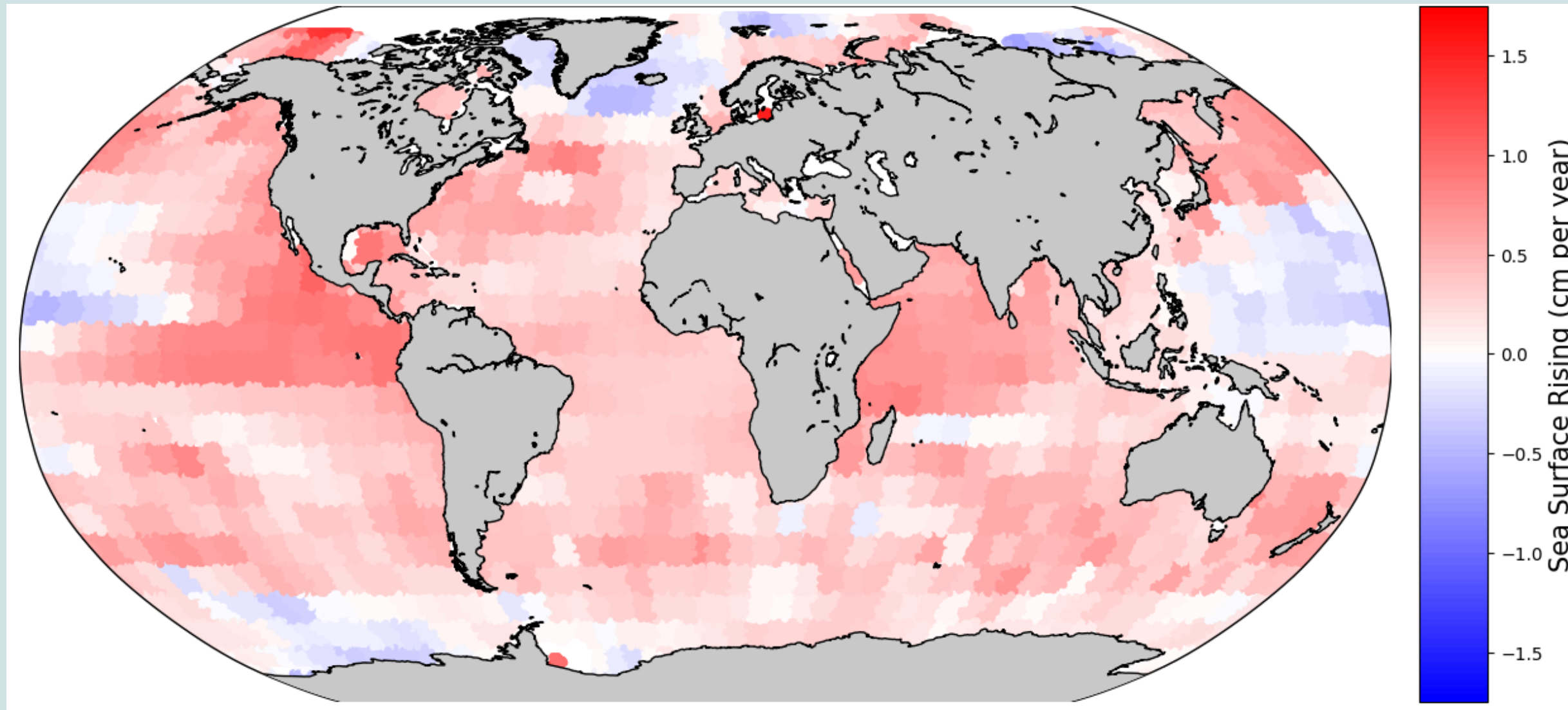
Comparison with the existing IPCC models



RCP Models:

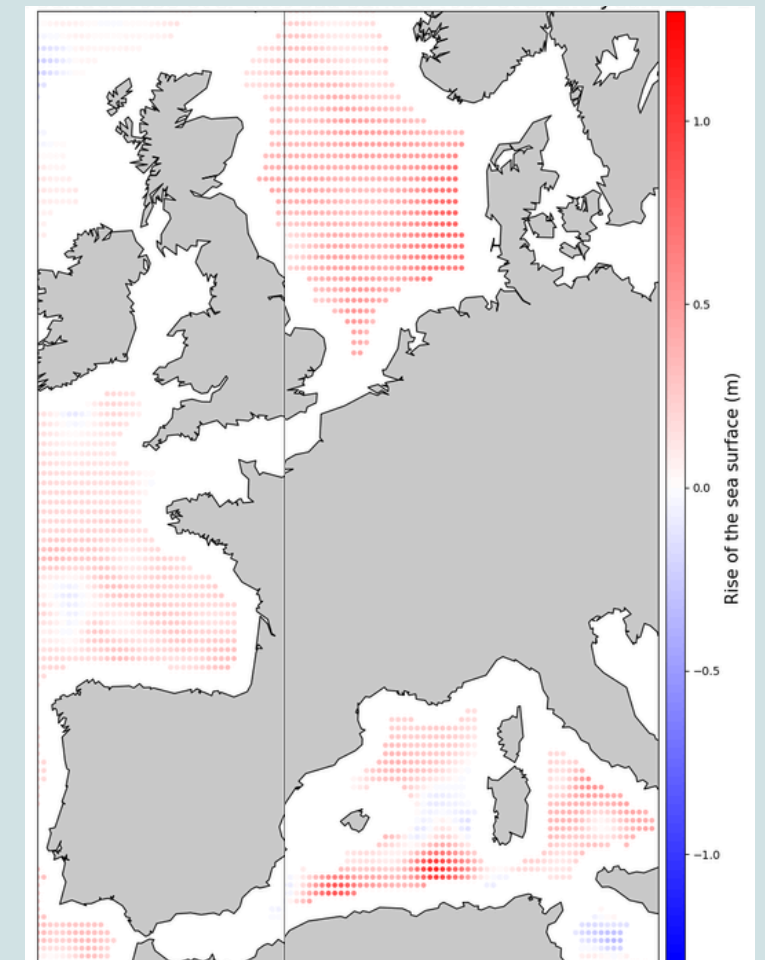
- Based on Radiative forcing
- Worst case: RCP8.5 $> 8.5 \text{ W.m}^{-2}$ by 2100
+3.2 to 5.4°C
- Best case: RCP2.6 peak at $\sim 3 \text{ W.m}^{-2}$
before 2100 and then decline
+0.9 to 2.4°C

How fast the sea level rose over the 2002-2015 period



- Computed on **groups of pixels** (~1000 pixels per group) to **avoid noise** in the data.
- The rise is calculated using the **slope** between the mean ssh of each group in 2002 and 2015.

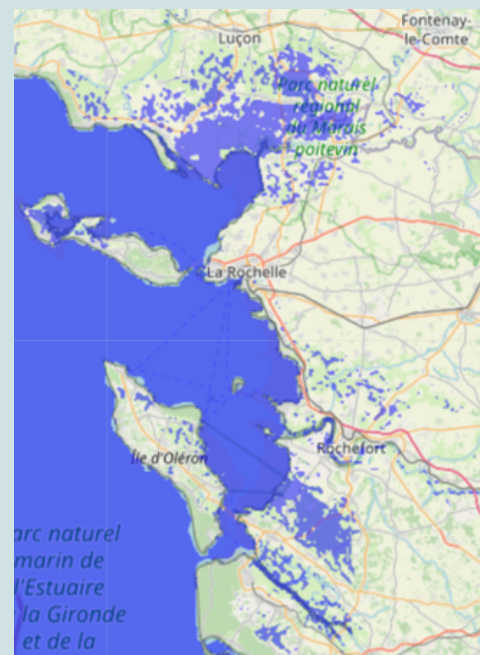
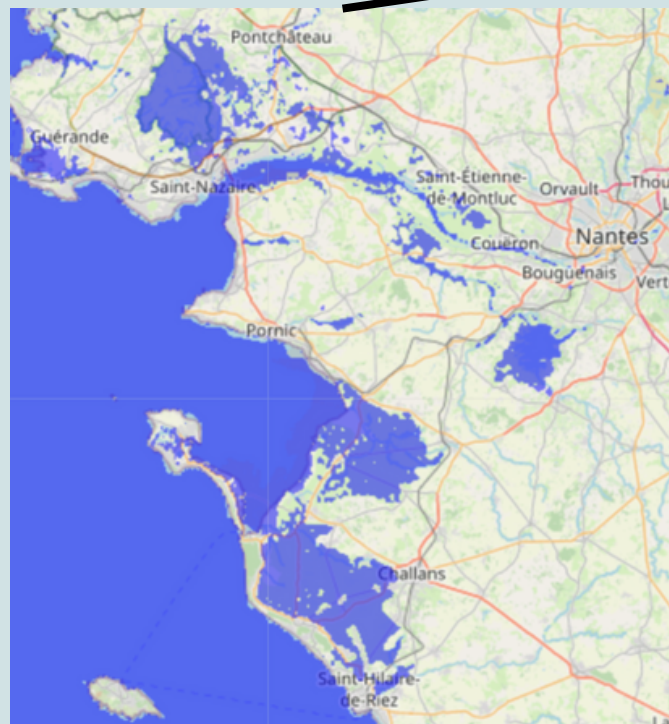
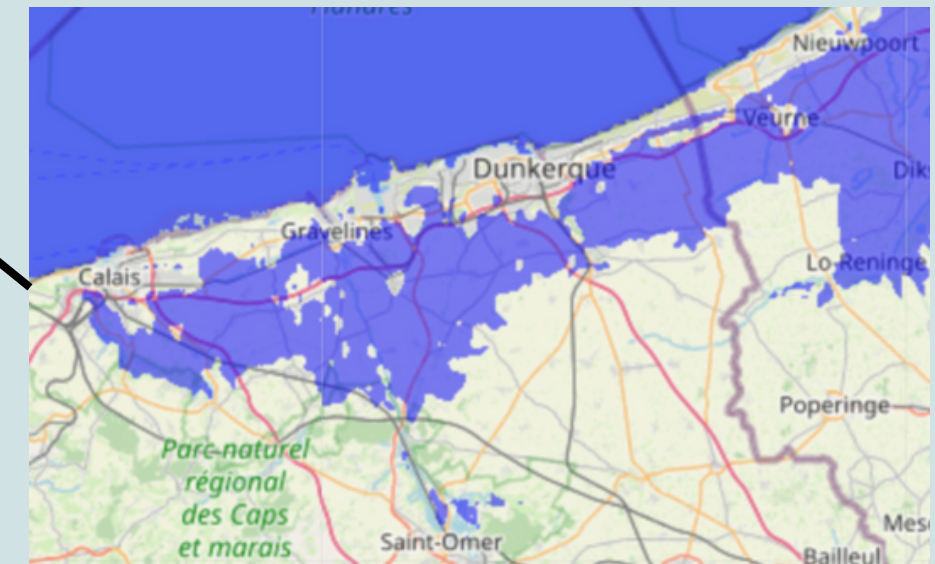
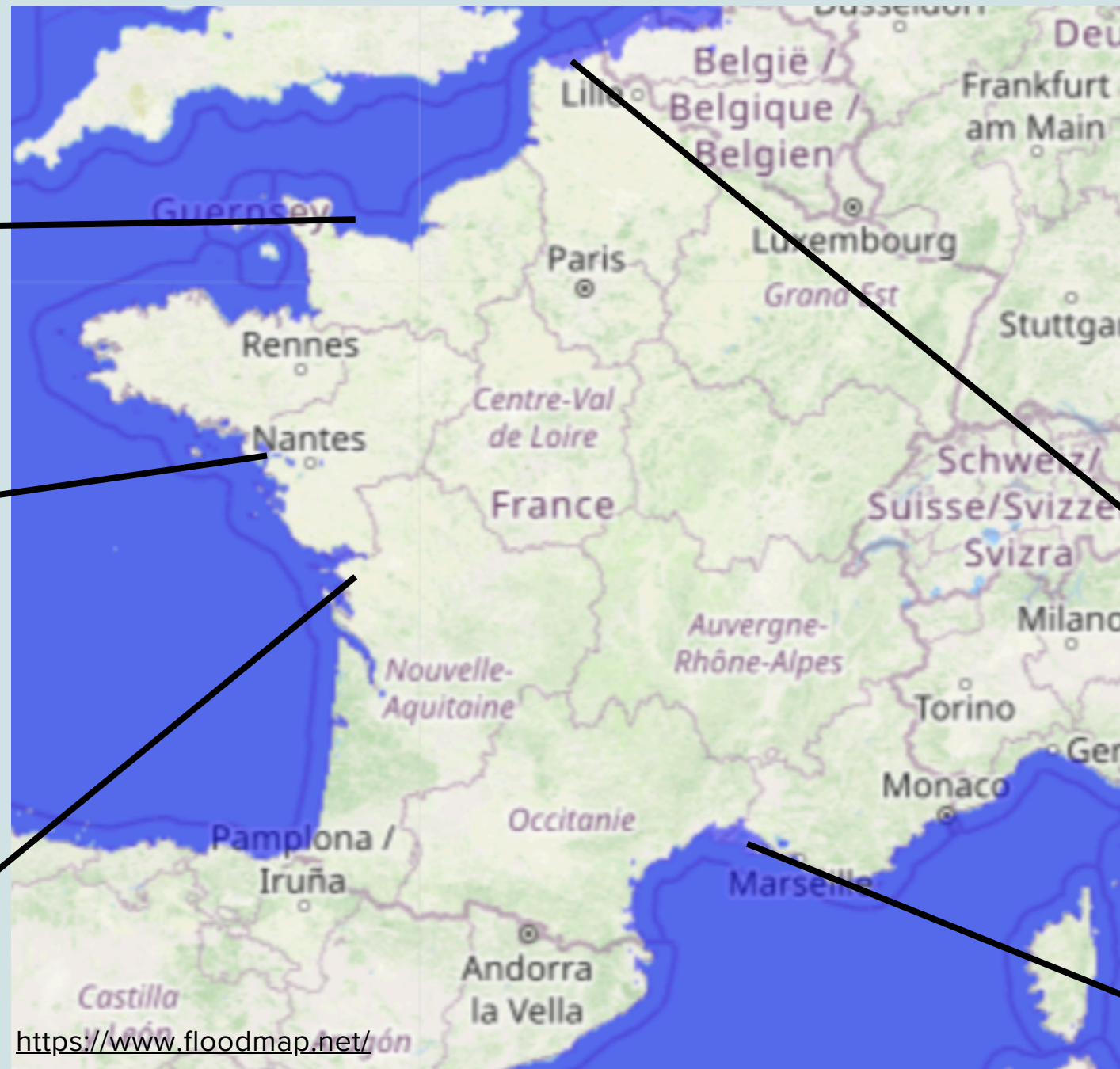
Linear projection for France in 2100



- As expected, it **globally rises** everywhere.
- However, there are **some decreases**: near the Poles and in the Pacific Ocean.
- Some areas where it **skyrockets** like near Mexico or in the north of the Indian Ocean.

Consequences

- Exponential prediction for **2100**.
- Rise of **1.2 m**.



**Thank you for your
attention**